Jayden Skelly | jls4rivers@gmail.com | *[Phone Redacted]* | Web
[Address Redacted]

EDUCATION

Montana State University: **M.S. in Biological Sciences** **Current**
Western Colorado University: **B.S. in Biology - Environmental Biology & Ecology** **2023**
Western Colorado University: **Geographic Information Systems (GIS) Certificate** **2023**
National Outdoor Leadership School: **Emergency Medical Technician Certificate** **2022**
University of Montana: **Field Study** **2021**
GPA: **Undergraduate: 3.75, Dean's List, Magna Cum Laude, Graduate: 3.7**

RESEARCH INTERESTS

I am dedicated to advancing conservation science by bridging quantitative skills with field-based research. I am interested in applying remote sensing, movement ecology, and spatial modeling to address large-scale ecological questions in collaborative settings. One area I would like to grow is integrating animal behavior with broader ecological processes such as demography, nutrition, and community ecology to inform robust, science-based conservation strategies.

RESEARCH EXPERIENCE

NSF Graduate Research Fellow August 2024 - Current
Montana State University, Bozeman, MT
In partnership with MT FWP and CIRA, developed a research project on pronghorn winter behavior and movement strategies. Analyzed 700 collared animals response to snow, forage, roads, and fences using habitat selection analyses, simulations, and time to event models. *Advised by Dr. Justine Becker, committee members: Dr. Sarah Bassing (MSU), Dr. Joshua Millspaugh (UM)*

USGS Ungulate Researcher (contracted) July 2026 - Current
US Geological Survey NOROCK, Bozeman, MT
Developed research on ungulate migration in Wyoming. Used tools such as camera traps, movement models, and literature reviews to support collaborators.

Research Technician - Research Affiliate April 2024 - August 2024
Yale University, New Haven, CT
Lead data collection for an international project near Uatumã Preserve, Amazonas, Brazil at a remote research station. Handled logistics, day to day plans, 2Tb+ of data management, and personnel. Work was centered around bioacoustics, biodiversity monitoring, and island biogeography.

USFWS Arctic Sea Duck Research Assistant June 2023 - July 2023
US Fish and Wildlife Service, Utqiagvik, AK
Assisted in research of the ESA listed Steller's and Spectacled Eider in remote arctic Alaska.

USGS CO-OP Ungulate Research Assistant May 2022 - Sept 2022
University of Wyoming, Jackson, WY
Assisted on nutrition, thermoregulation, and disease dynamic projects on bighorn sheep, moose, and mule deer. Captured lambs in mountainous terrain, performed necropsies, botanized, on backpacking trips.

Remote Sensing Thesis Research January 2021 - May 2022
Western Colorado University, Gunnison, CO
Collaborated with the BLM on beetle kill winter surveys paired with LandSat derived vegetation metrics.

Golden Eagle Toxicology Research Assistant December 2020 - June 2021
HawkWatch International, Salt Lake City, UT
Access to nest sites for bloodwork and installation of gps-loggers performed with high-angle rope access.

OTHER TECHNICIAN WORK

River Otter Surveyor, Western Colorado University, Colorado. 2021.
Northern Goshawk Nestling Technician, Utah State University, Utah. 2020.
Black Hills National Forest Internship, Student Conservation Association, South Dakota. 2019.

PROFESSIONAL EXPERIENCE

XPRIZE Rainforest Finalist - *Yale University, Rio Negro, Brazil*
Competed in the finals of the XPRIZE Rainforest 5-year $10 million dollar biodiversity startup competition on the Map of Life Rapid Assessments Team (2024).

BLM GIS Intern - *Bureau of Land Management, Lakewood, CO*
Provided GIS support for the Gunnison Field Office (2023) and BLM Colorado State Office (2024).

Ski Patroller - *Arapahoe Basin Ski Area, Dillon, CO*
Worked as an EMT, performed medicine in an austere environment, kept track of patroller locations around the mountain, dispatched reports from emergency numbers.

Mountain Rescue Trainer and Responder - *Western Mountain Rescue Team, Gunnison, CO*
Volunteer responder on the primary search and rescue team of Gunnison County. Responded to calls with some missions reaching over 24hrs in length. Insertion to the scene occurred by foot, ski, climbing, mountaineering, helicopters, and snowmobile/ATV. Served on the board as the Field Trainer (2021) and Social Media-Marketing Coordinator (2020).

FELLOWSHIPS AND AWARDS

MSU Crawford Habitat Scholarship, 2026-2027 ($10,000)
MSU Crawford Habitat Scholarship, 2025-2026 ($10,000)
MontanaView Remote Sensing Award, 2025 ($1,000)
XPRIZE Rainforest 2nd Place, 2024 ($2,000,000 awarded to team)
National Science Foundation Graduate Research Fellowship Program, 2024-2029 ($159,000)
Western Colorado University, Casey McKenney and Matt Noddin Biology Award, 2021
Western Mountain Rescue, Paul Maniachi Avalanche Memorial Education Award, 2021
Western Colorado University, Ridgeline Merit Award, 2019 - 2023

PUBLICATIONS AND PRESENTATIONS

"Forage, snow, and linear features shape pronghorn winter habitat selection and access " *The Wildlife Society Annual Conference.* Des Moines, Iowa. November 2026.

"Forage, snow, and linear features shape pronghorn winter habitat selection and access " *WAFWA 31st Biennial Pronghorn Workshop.* Colorado Springs, Colorado. August 2026.

"Dynamic snow properties, vegetation access, and fencing shape pronghorn winter habitat selection across Montana" *The Wildlife Society, Montana Chapter, Annual Conference,* Missoula, Montana. February 2026.

"Examining the Environmental Characteristics that Shape Winter Pronghorn Movement" *The Wildlife Society Annual Conference,* Edmonton, Canada. October 2025. *(poster.)*

"From sagebrush to simulations, unravelling pronghorn winter movement" *Thursday Science Seminars, Montana State University,* Bozeman, Montana. May 2025.

"Winter wandering: pronghorn strategies for surviving severe winters" *Thursday Science Seminars, Montana State University,* Bozeman, Montana. April 2025.

"Pine squirrels and remote sensing in the Powderhorn" *Western Colorado University Science Seminars*, Gunnison, Colorado. November 2022.

Skelly, Jayden. Lake San Cristobal Priority Fuels Treatment GIS Development. *Bureau of Land Management.* 2023.

Skelly, Jayden. Tamiasciurus hudsonicus Presence-Absence in a Forest with Insect-Elicited Mortality. *Western Colorado University*. 2022. *(undergraduate thesis)*

OUTREACH

- Montana State University ecology department seminar committee vice president, 2026
- Mentored 1x undergraduate in developing research proposal for submission, 2026 *(awarded)*
- Conceptualized and maintain a 300+ enrolled ecology job board alert system
- Ecological Society of America Member / The Wildlife Society Member *(current)*

SKILLS

Fieldwork Based

- Attended 3-day wildlife chemical immobilization course, administered sedatives, 2025
- Attended 1-day Field Futures training on inclusive fieldwork, 2025
- Active NR-EMT with professional experience in backcountry medicine
- Blood draw, morphometrics, and gps-logger / collar / banding installation on ungulates and raptors
- Managed fieldwork operations, day to day logistics, and personnel
- Driven ATV/vehicles in 4wD, flying in helicopters, riding in boats, snowcats or on snowmobiles
- Worked in remote, mountainous environments, or international locations
- Performed necropsies, habitat assessments, used ARU's, camera trapping, eDNA, botany

Computer Based

- Advanced spatial analysis skills in R, Python, Google Earth Engine, ESRI suite
- Advanced understanding of remote sensing data sources, spectral bands, sensor types
- Parallel computing in HPC context with Linux, Slurm, Globus
- Applied Machine Learning (CNNs, RFs) to data, worked with pretrained models such as BirdNet

Statistical Based

- Attended 2-day workshop with Dr. Brian Smith on fitting GAMs to iSSAs, 2026
- Attended 1-day SECR (Spatially Explicit Mark Recap) class by Dr. Beth Gardner (UW), 2025
- Frequentist statistics including study design, inference, hypothesis testing with likelihoods
- Extensive work on habitat selection analyses, including validation with simulation, cross fold
- Used animal movement analysis techniques ie. (`amt, ctmm, marcher, momentuHMM,` etc.)
- Familiar with mark-recap, occupancy models, and other population estimations
- Learning and willingness to learn Bayesian approaches

Conservation and Policy Based

- Collaborated with state wildlife agencies (MT,WY) on developing projects for management
- Collaborated with federal agency (BLM) on applied timber management research
- Interested in the integration of research with science communication
- Assisted with state level lobbying for environmental topics

SELECTED MEDIA COVERAGE

Kidangoor, Abhishyant. "At the 'Biodiversity...democratize rainforest tech" *Mongabay.* 2024.
O'Donoghue, J. Amy. "Saving the golden eagle." *Deseret News.* 2021.
O'Donoghue, J. Amy. "Golden eagle…struggling …the West." *Associated Press*. 2021.
Heaps, Spenser. "Capturing golden eagles. Here's how I got the pictures." *Deseret News.* 2021.

REFERENCES

[References Redacted]